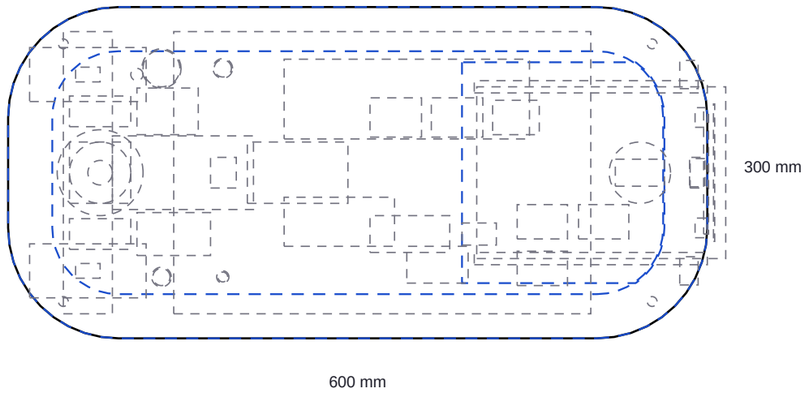
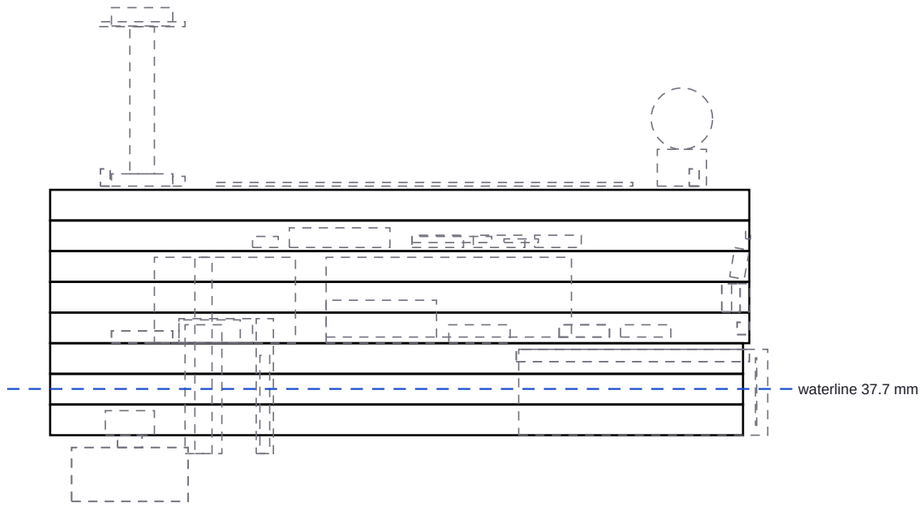
VARUNA AI Pod

600 x 300 mm thermocol boat | Your design, upgraded

PLAN (from above), scale 1:5



SIDE, scale 1:5



Size	600 x 300 mm, hull 200 mm tall, 348 mm to GPS top
Weight	4184 g empty, 4565 g with 381 g of plastic
Draft	37.7 mm empty, 40.9 mm full
Reserve buoyancy	10.5 kg before the deck goes under
Stability (GM)	104 mm: very stable
Trim	bow 2.6 mm higher than stern empty; slide the battery to level
Collection	1.9 L pull-out drawer
Camera	102 mm above water, tilted 10 deg down
Battery	Li-ion 3S6P 20 Ah (222 Wh), about 4.4 h
Thermocol	8 sheets 1000 x 500, or 3 sheets 2 x 3 ft (610 x 915)

How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.

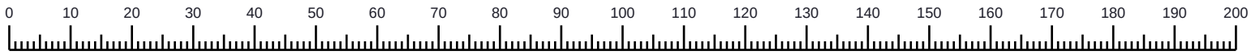
cut line

cut out (window / pocket)

fold / glue line

placement guide, do not cut

skewer hole: lines up the layers



ruler in mm: the 0 to 200 marks must be exactly 200 mm apart

100 mm x 100 mm

How to print on A3 so every part comes out real size

1. Paper: A3, landscape. Printer: 'A3' selected as the paper size.
2. Scaling: 'Actual size' or '100%'. Turn OFF 'Fit to page', 'Shrink oversized pages' and 'Fit to printable area'. On a phone, choose 'Original size' / 'Do not scale'.
3. Print this page only. Measure the square with a steel ruler: both sides must be 100 mm, and the ruler marks must match your ruler. 1 mm off is fine; more is not.
4. If it is wrong, fix the printer setting and print this page again. Do not go on.
5. When it is right, print all the other pages with the SAME settings.
6. Big parts are split across pages (tiles A1, A2, B1...). Cut each tile on its border, lay the next tile over the grey overlap strip, line up the grey 50 mm grid and the measurement ruler, and tape it. Check the long measurement with a tape before cutting.
7. Glue-stick the paper on the thermocol, cut along the solid black line, cut out the red parts, and push a skewer through every + mark.

Line types:

- cut line
- cut out (window / pocket)
- fold / glue line
- placement guide, do not cut
- skewer hole: lines up the layers
- measurement (mm), do not cut

A1

32 dia: 125 from stern, 85 left

Turbidity

16 dia: 175 from stern, 85 left

TDS

slot starts 45 from stern

0 50 100 150 200 250 300 350 400

270 overall width

230 beam slot

40

overlap: tile A2 goes on top

pH

16 dia: 125 from stern, 85 right

Temperature

10 dia: 175 from stern, 85 right

570 overall length

BASE layer 1

A2

DRAWER TUNNEL (cut away)

400 450 500 550

150 tunnel width

190 tunnel, from bow edge

<- must measure 50 mm

A1

32 dia: 125 from stern, 85 left

Turbidity

16 dia: 175 from stern, 85 left

TDS

0 50 100 150 200 250 300 350 400

270 overall width

overlap: tile A2 goes on top

pH

16 dia: 125 from stern, 85 right

Temperature

10 dia: 175 from stern, 85 right

570 overall length

BASE layer 2

<- must measure 50 mm

A2

DRAWER TUNNEL (cut away)

400 450 500 550

150 tunnel width

190 tunnel, from bow edge

A1

32 dia: 125 from stern, 85 left

Turbidity

16 dia: 175 from stern, 85 left

TDS

ESC 30 A bidirectional

2-channel relay module

Li-ion 356P 20 Ah batt

0 50 100 150 200 250 300 350 400

270 overall width

overlap: tile A2 goes on top

ESC 30 A bidirectional

12 V to 5 V 5 A buck c

Spray pump 12 V

CN3791 solar charge mo

pH

16 dia: 125 from stern, 85 right

Temperature

10 dia: 175 from stern, 85 right

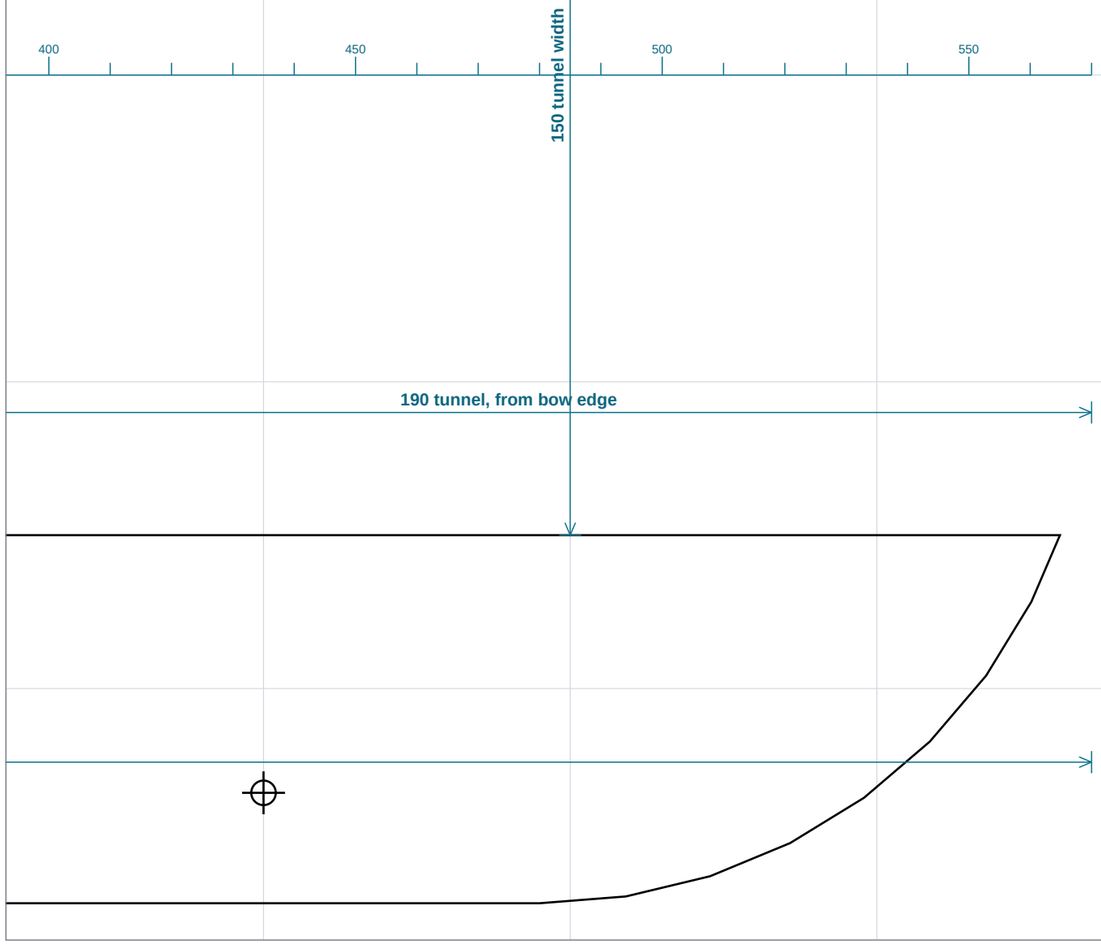
570 overall length

BASE layer 3 - top face is the lower-bay floor

<- must measure 50 mm

A2

DRAWER TUNNEL (cut away)



A1

R90 outside

R55 inside

35 wall

0 50 100 150 200 250 300 350 400

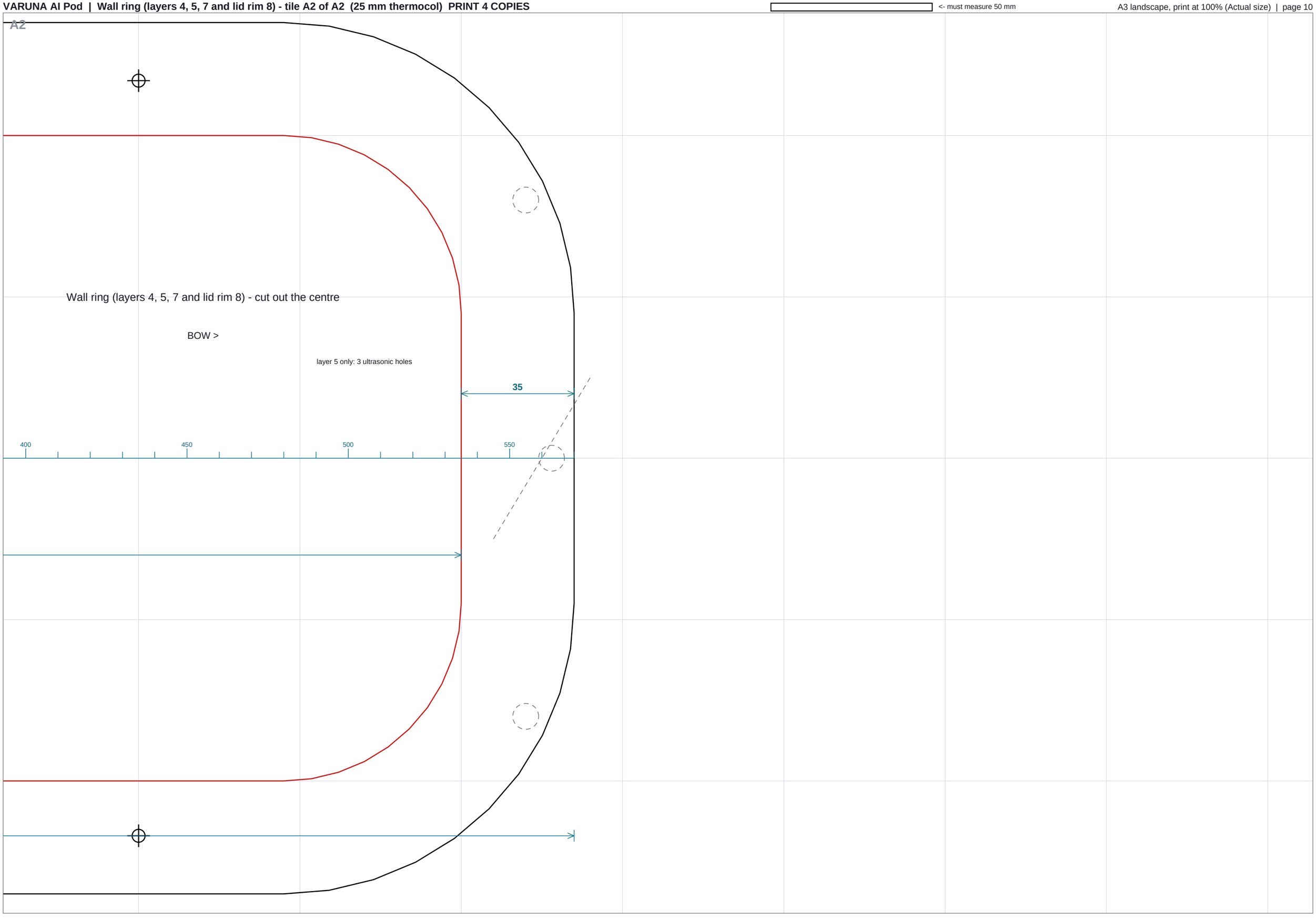
270 overall width

200 inside

overlap: tile A2 goes on top

500 inside

570 overall length


A2
Wall ring (layers 4, 5, 7 and lid rim 8) - cut out the centre
BOW >
layer 5 only: 3 ultrasonic holes
35
400
450
500
550

<- must measure 50 mm

A1

R90 outside

R45 inside

45 wall

270 overall width

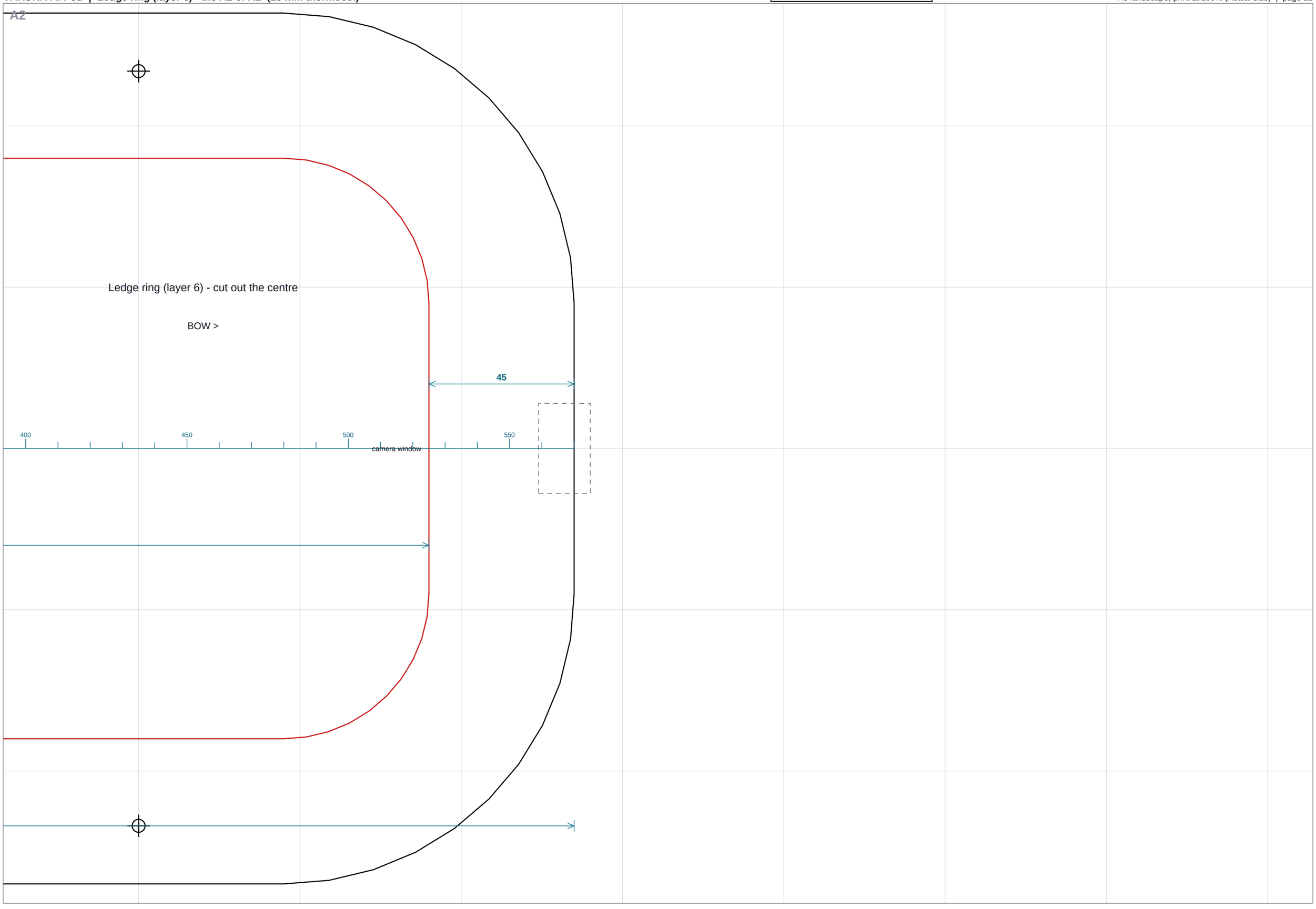
180 inside

480 inside

570 overall length

overlap: tile A2 goes on top

<- must measure 50 mm



<- must measure 50 mm

A1

nozzle holes 6 dia: 45 and 525 from stern, 105 each side

nozzle

status LED

5 dia, 105 from stern, 80 left

GPS mast

solar panel 340 x 230

270 overall width

overlap: tile A2 goes on top

panel cable 8 dia, 305 from stern

mast hole 21 dia, 75 from stern

0 50 100 150 200 250 300 350 400

570 overall length

A2

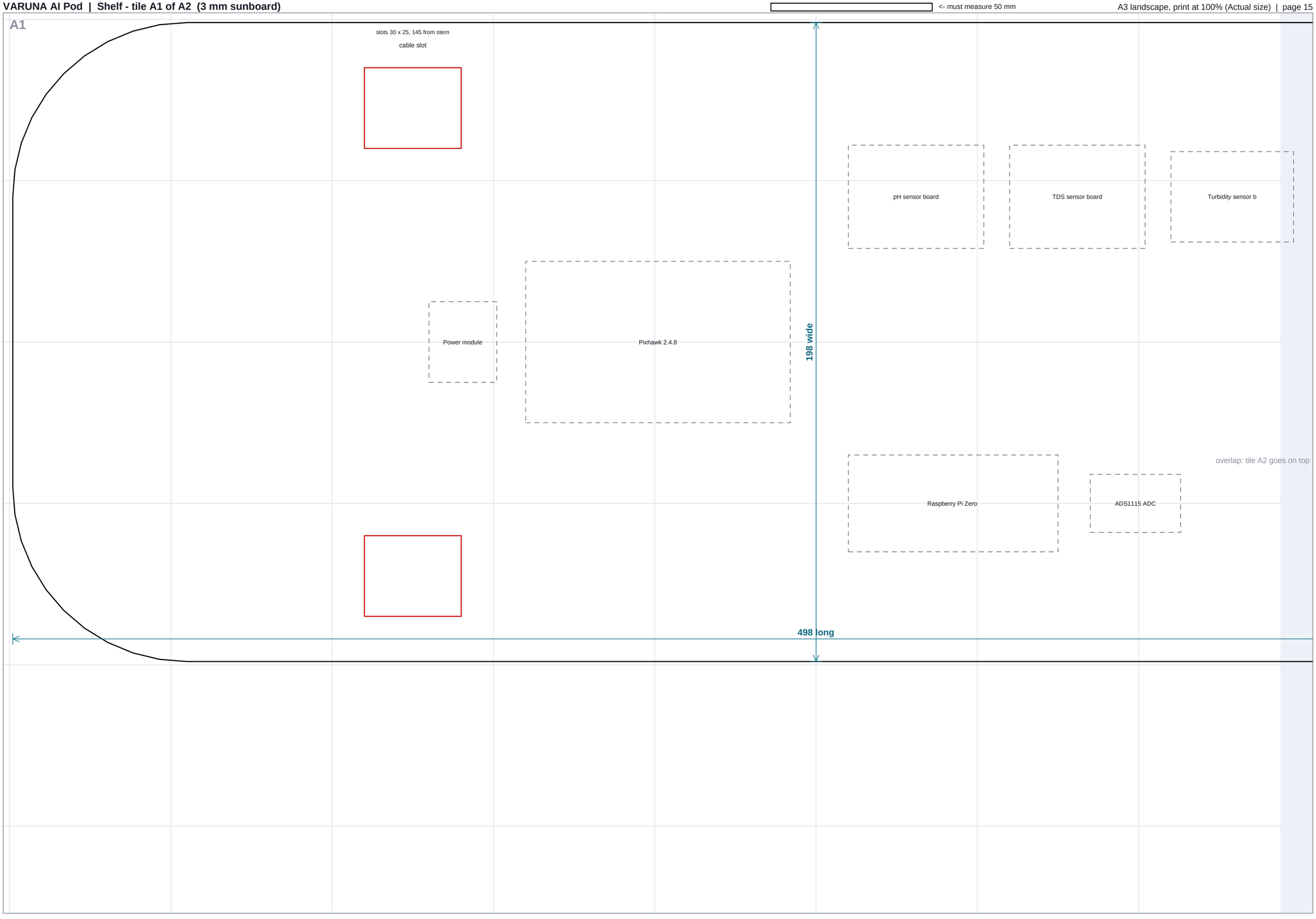
nozzle

camera cable hole 20 dia, 515 from stern

servo plate

BOW >

400 450 500 550


A1
slots 30 x 25, 145 from stern
cable slot
pH sensor board
TDS sensor board
Turbidity sensor b
Power module
Pixhawk 2.4.8
198 wide
overlap: tile A2 goes on top
Raspberry Pi Zero
ADS1115 ADC
498 long

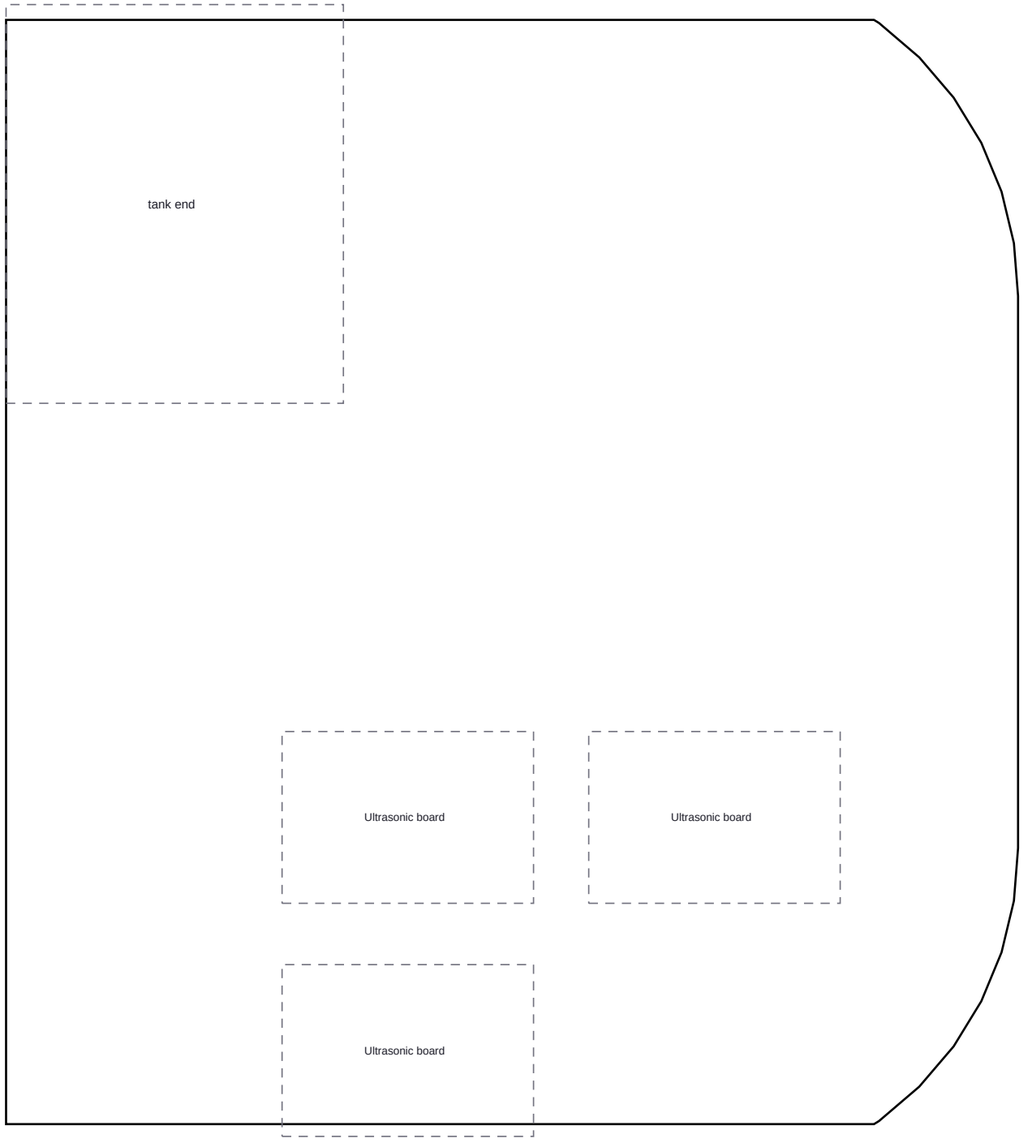
A2

BOW >

<- must measure 50 mm

Drawer tunnel roof

5 mm sunboard | 165 x 184 mm

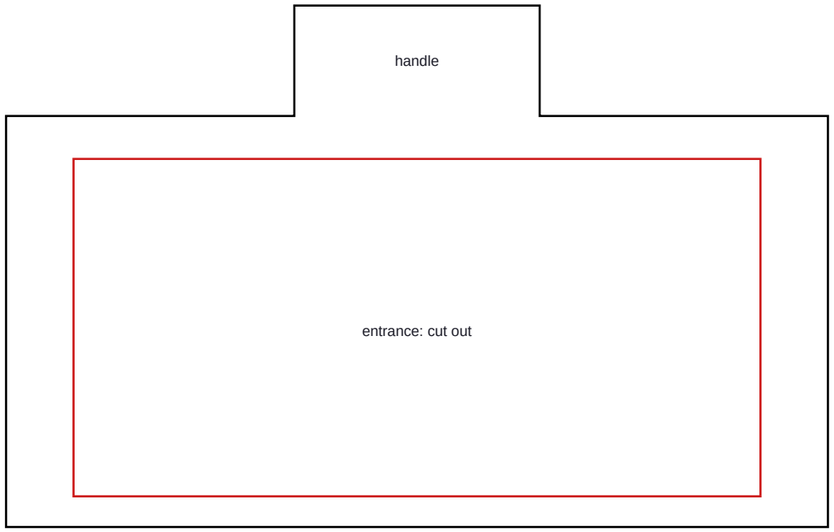


Drawer bottom

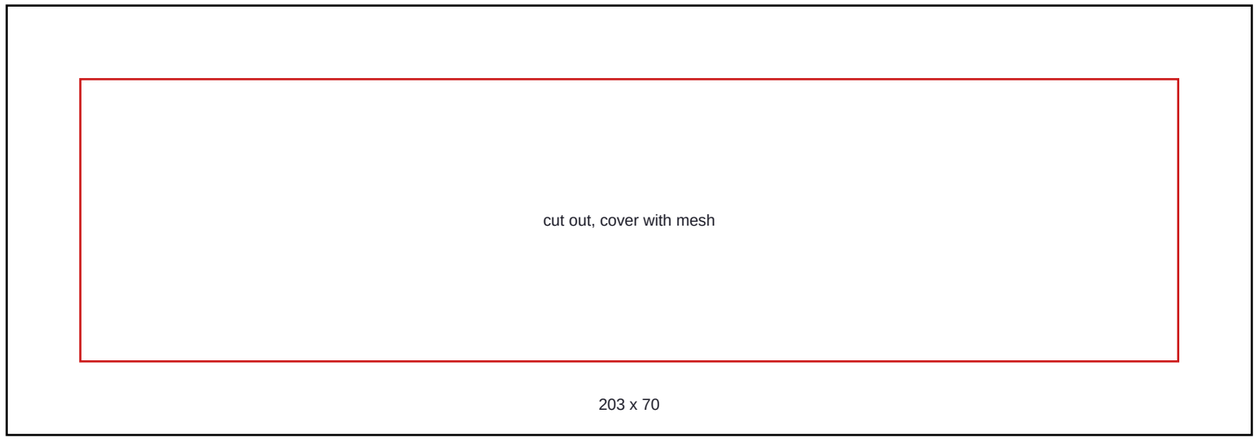
3 mm sunboard | 203 x 134 mm



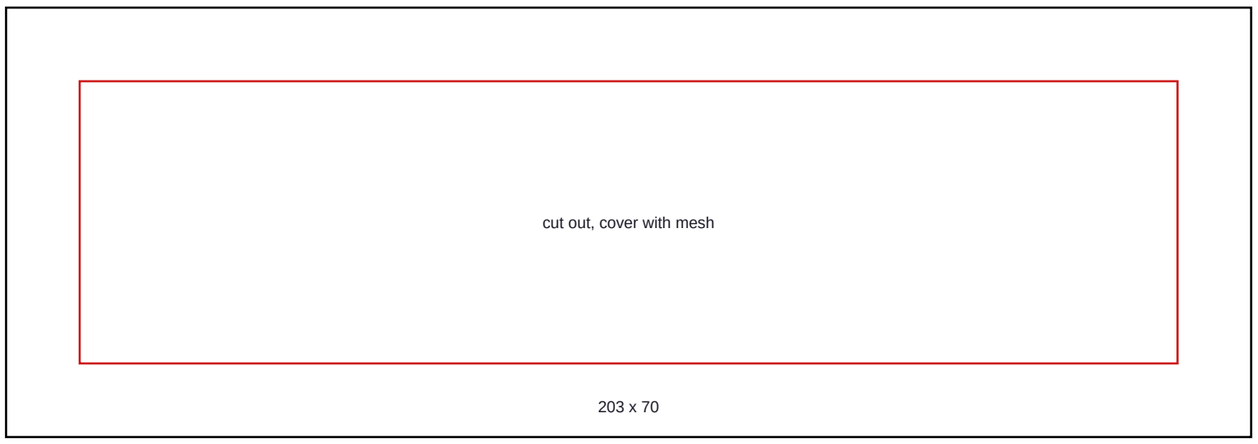
Drawer front (entrance + handle)
3 mm sunboard | 134 x 85 mm



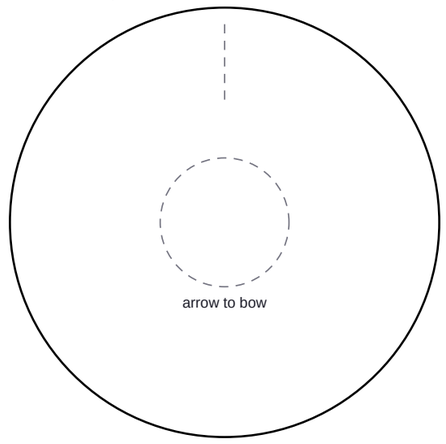
Drawer side (left)
3 mm sunboard | 203 x 70 mm



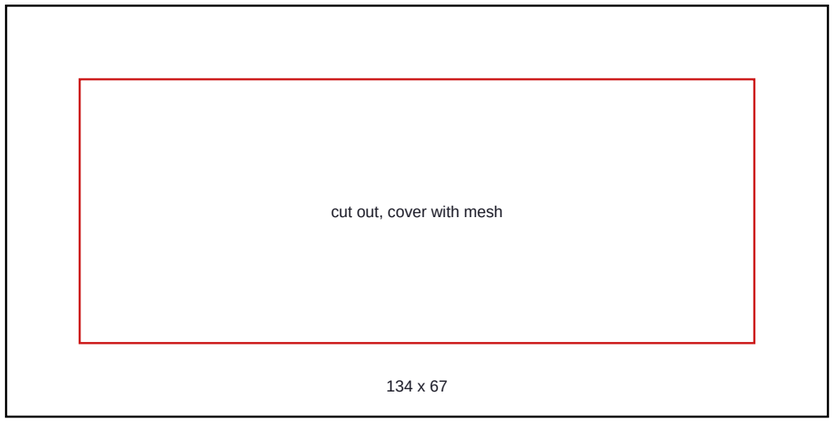
Drawer side (right)
3 mm sunboard | 203 x 70 mm



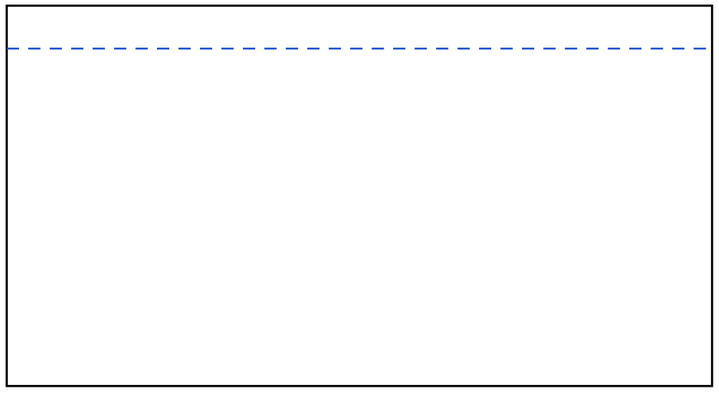
GPS disc
5 mm sunboard | 70 x 70 mm



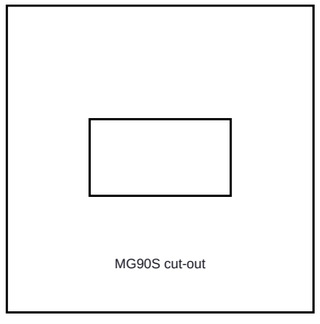
Drawer back
3 mm sunboard | 134 x 67 mm



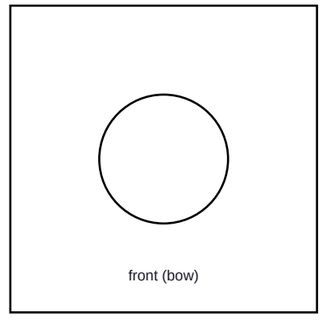
One-way flap
PET sheet | 115 x 62 mm



Servo plate
3 mm sunboard | 50 x 50 mm



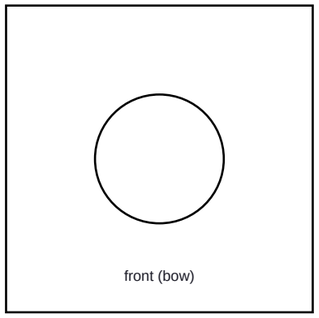
Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



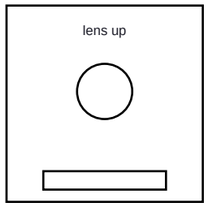
VARUNA AI Pod | small parts 3 of 3

<- must measure 50 mm

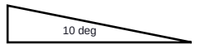
Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



Camera plate
3 mm sunboard | 32 x 32 mm



Camera wedge side (10 deg)
3 mm sunboard | 30 x 6 mm



Camera wedge side (10 deg)
3 mm sunboard | 30 x 6 mm

